

# Phytochemicals

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# What is Phytochemicals

- ❖ All plants — including fruit, vegetables, beans and grains — produce phytochemicals. They are part of the plant's immune system and help protect the plant from viruses, bacteria, fungi and parasites.
- ❖ Phytochemicals can offer humans some of that same protection. While some phytochemicals are also vitamins, they aren't essential to our health like vitamins and minerals. Instead, they protect your cells from damage caused by environmental toxins and the body's natural chemical (metabolic) processes. "Phytochemicals have antioxidant properties and offer protection that decreases the risk of many diseases."

# Health Benefits Of Phytochemicals

- ❖ Phytochemicals are substances found in plants that may have health benefits when consumed. Some potential benefits of phytochemicals include:
  - Immune system: Phytochemicals can help balance the immune system, which can protect against infection and disease.
  - Inflammation: Phytochemicals can reduce inflammation ,which is associated with inflammatory diseases.
  - Cancer: Phytochemicals may help slow the growth of cancer cells, prevent damaged cells from reproducing, and protect cells and DNA from damage.
  - Hormones: Phytochemicals may help regulate hormones.
  - Antimicrobial: Phytochemicals can work as antimicrobial agents to reduce the chance of viruses and bacteria growing in the body.
- ❖ Some examples of phytochemicals include:
  - Isoflavones: Found in legumes, isoflavones may reduce the risk of breast and prostate cancers.
  - Ellagic acid: Found in raspberries, strawberries, blackberries, grapes, pomegranates, walnuts, and pecans, ellagic acid may reduce cancer risk and lower cholesterol.
  - Phytosterols: Phytosterols may support prostate health, hair growth, and reduce LDL cholesterol.
- ❖ The health benefits of phytochemicals depend on their purity and structural stability.

# Types of Phytochemicals

## ❖Types of Phytochemicals:

- Anthocyanidins produced by red and purple berries.
- Beta-carotene found in orange and dark green leafy vegetables.
- Catechins present in black grapes, apricots strawberries.
- Carotenoids produced in pumpkin, carrots and bell peppers.
- Flavonoids found in tea and wine.
- Isoflavones contained in soya beans.
- Polyphenols produced by cloves, berries dark chocolate.

# Foods Rich in Phytochemicals

❖ Phytochemicals are found in all plant-based foods including:

- Fruits
- Vegetables
- Legumes
- Grains
- Beans
- Nuts
- Seeds
- Herbs and spices

❖ The best way to get a variety of phytochemicals is to eat many different kinds of fruits and vegetables. Keep in mind, the different colors of fruits and vegetables represent different phytochemicals. So aim for a mixture of colors and to fill half your plate with fruits and vegetables at each meal."

# Spices and Health Benefits

- ❖ Herbs and spices have been widely valued for centuries, both for their healing properties and their flavor. Modern research has found evidence for numerous health benefits.
- The World Health Organization (WHO) defines health as a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.
- Explanation: The World Health Organization (WHO) says that health means being completely well in your body, mind, and social life, not just being free from illness or problems. So, it's about feeling good all around, not just avoiding sickness.

# Cardamom

- ❖ Antioxidant and cancer fighting properties: This flavourful spice is packed with health benefits and has been used in traditional medicine for centuries. Some evidence shows that this herb has antioxidant properties and has the ability to produce compounds that may help fight cancer cells.
- ❖ Anti-inflammatory effects: The antioxidants that are found abundantly in cardamom protect cells from damage and stop inflammation from occurring.
- ❖ Treats bad breath and prevents cavities: Cardamom has long been used to freshen breath and improve oral health. This is related to its ability to fight common mouth bacteria. It is used in karanji (a small pastry pocket stuffed with poppy seeds, grated coconut, sugar, nuts and cardamom) and mithai (an assortment of Indian sweets)."

# Cumin

- ❖ Aids in digestion and reduce food-borne infections: It is rich in antioxidants, and has been shown to exhibit anti-microbial and anti-inflammatory properties. Some studies have demonstrated that cumin can also aid in digestion and help in reducing food-borne infections. Some research also suggests that cumin powder, when added to a low-calorie diet, can help with weight loss.
- ❖ Helps manage blood sugar levels: Cumin might lower blood sugar levels in some people with diabetes. However, other human studies have shown mixed results and therefore more research is necessary to confirm the benefits of cumin seeds for those with diabetes. It is used in a popular Indian snack called murukku (deep fried coils of rice flour and spices). This snack should be enjoyed in moderation, though, as it is high in fat and may lead to weight gain."

# Ginger

- ❖ "Reduces nausea and inflammation: Ginger has long been associated with reducing nausea, pain and inflammation. It is especially effective when it comes to pregnancy-related nausea such as morning sickness. Additionally, it has also been found to be equally effective as drugs against menstrual pain. Much of ginger's beneficial properties are attributed to gingerol, which has powerful anti-inflammatory and antioxidant effects.
- ❖ Helps with osteoarthritis: People who used ginger to treat their osteoarthritis saw significant reductions in pain and stiffness, especially in osteoarthritis of the knee. Some people, however, may not enjoy the taste of ginger and may experience stomach upset.
- ❖ Lower blood sugars and improve heart health: Recent studies have found that ginger may have anti-diabetic properties. A study found that ginger dramatically improved HbA1c levels as well as other biochemical markers that are indicative of risk factors of heart disease. However, more studies maybe needed in humans to validate this claim.
- ❖ Ginger is an essential ingredient in Indian cuisine, and can be found in dishes like chickpea stew, aloo gobi (a dish of potatoes and cauliflower), matar paneer (cheese and peas in tomato sauce), dal makhani (lentils cooked with butter) and more."

# Turmeric

- ❖ Medicinal properties that may prevent heart attacks. It has been used in India for thousands of years as a spice and medicinal herb. Research has shown that it contains compounds with medicinal properties, the most important of which is curcumin, the main active ingredient in turmeric. Preliminary studies found that curcuminoids from turmeric may reduce the number of heart attacks patients have after bypass surgery.
- ❖ Boosts brain function: Curcumin in turmeric has beneficial effect on levels of brain-derived neurotrophic factor (BDNF), which has been associated to delay or even reverse many brain diseases and age-related decreases in brain function.
- ❖ Shows benefits against depression: A small study has shown that curcumin can help with patients suffering from depression. This could also be linked to its effects in boosting BDNF levels, which is implicated in depression. There is also some evidence that curcumin can boost brain neurotransmitters, including serotonin and dopamine.
- ❖ Turmeric is also used as a dietary supplement to treat inflammatory arthritis, as well as stomach, skin, liver and gall bladder problems."

# Coriander

- ❖ Anti-microbial and anti-fungal properties. This superfood claims to have anti-microbial and anti-fungal properties, and is known to help lower blood pressure, blood sugar levels and cholesterol levels.
- ❖ Helps lower blood sugar: Coriander seeds, extract, and oils may all help lower blood sugar levels by increasing insulin release from the beta cells of the pancreas. People with low blood sugar or those taking diabetes medication should use it with caution.
- ❖ You are likely to taste coriander in aloo tikki (the Indian version of croquette or hash brown) or samosa (a fried or baked triangular puff filled with potatoes, onions, peas or lentils)."

# Cinnamon

- ❖ Helps to reduce cholesterol levels: A study comparing 26 spices placed cinnamon at the top for its high amount of antioxidants, surpassing garlic and oregano. Its anti-inflammatory properties may prevent the formation of free radicals that damage your cells and nervous system, and also help to reduce cholesterol levels.
- ❖ High in antioxidants: Cinnamon contains high levels of antioxidants, such as polyphenols that can protect the body from oxidative damage caused by free radicals. These antioxidants have potent anti-inflammatory properties, which may help your body fight infections and repair tissue damage.
- ❖ Helps to reduce risk of heart disease: Cinnamon has been shown to reduce levels of total cholesterol, bad cholesterol (LDL) and triglycerides, while maintaining levels of good cholesterol (HDL). The special compounds in cinnamon may help reduce the risk of heart disease. It is often used in keema (an Indian spiced lamb), chicken dhansak (an Indian curry) and namkeens (Indian savoury snacks)."

Thank You